



CAPSTONE PROJECT -

CLOUD-NATIVE OMNI-
CHANNEL RETAIL ANALYTICS
PLATFORM

LAKSHYA S

THE BUSINESS PROBLEM

- The Scenario: Modern retail (like Amazon or Flipkart) operates across stores, commerce websites, mobile applications, third party application.
- The Point: Data is currently siloed, inconsistent, and delayed.
- The Impact: Decision-makers can't tell which products perform best across all channels or if stock levels match real-time demand.
- The Goal: Unify this "data chaos" into a single source of truth for Merchandising, Marketing, and Operations teams



PROJECT OBJECTIVES

HEXAWARE

- Build an automated, incremental ingestion pipeline.
- Implement a Medallion Architecture (Bronze/Silver/Gold) using Delta Lake.
- Ensure "Production-Ready" code with CI/CD and environment separation

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ASSUMPTIONS

Assumed that a "Negative Inventory" record (where $\text{stock_on_hand} < 0$) is a system sync error – purchase in store and online (same second)

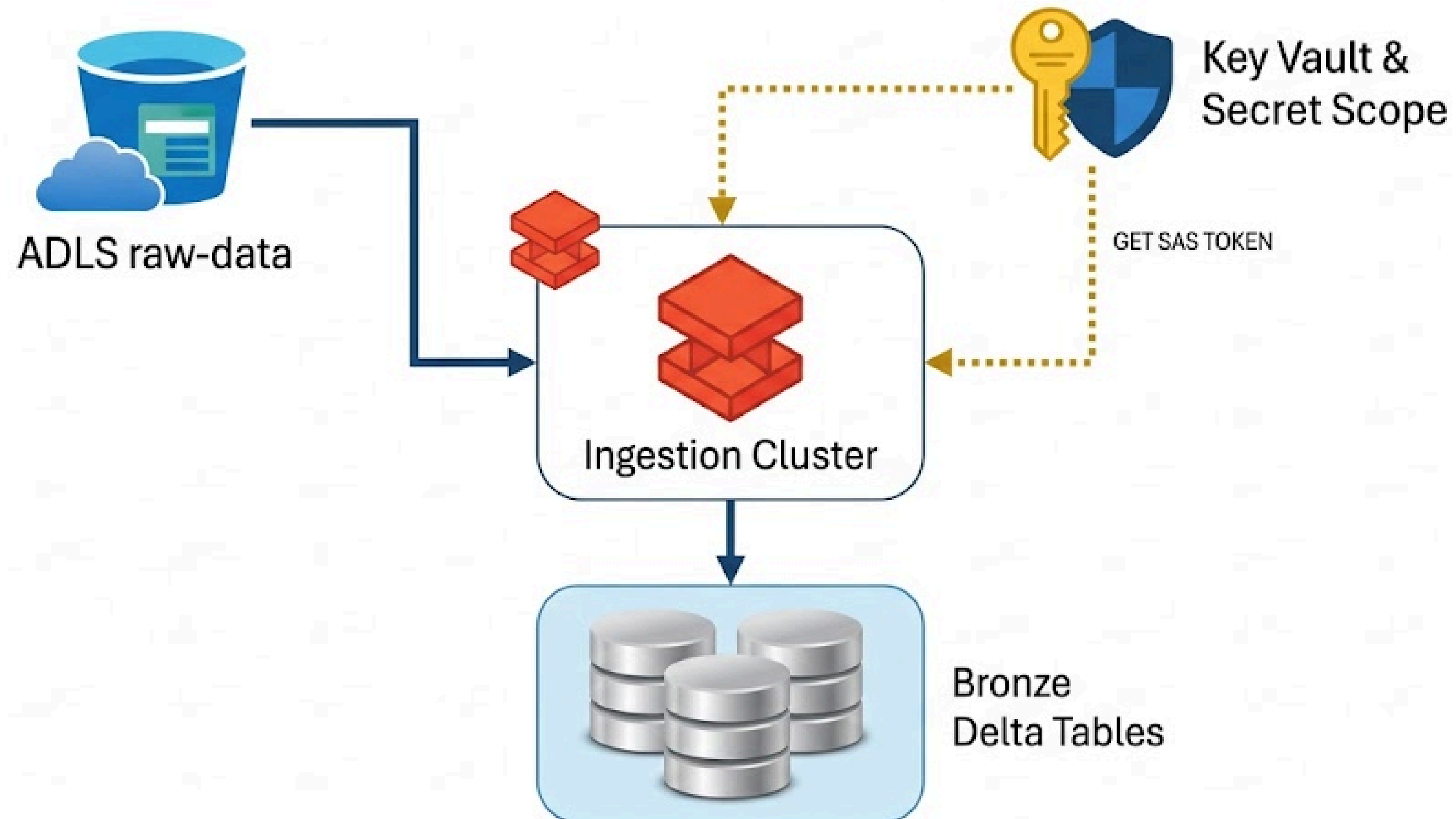
Assumed all order_time data should be converted to a single Standard Timezone (e.g., UTC) for global reporting

Assumed that a "New Customer" becomes a "Repeat Customer" only after their 3rd purchase

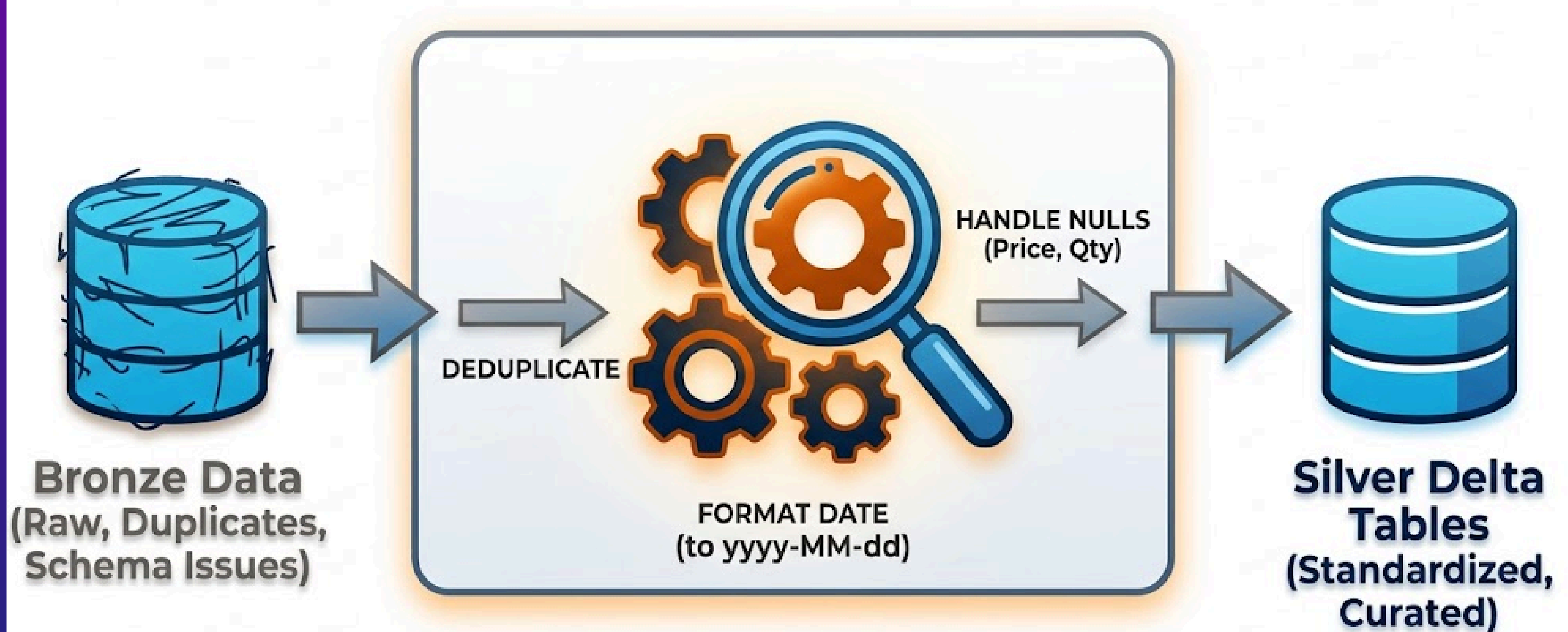
TECHSTACK

INCONSISTENCIES

SECURE INGESTION PATH: SAS TOKEN RETRIEVAL VIA DBUTILS



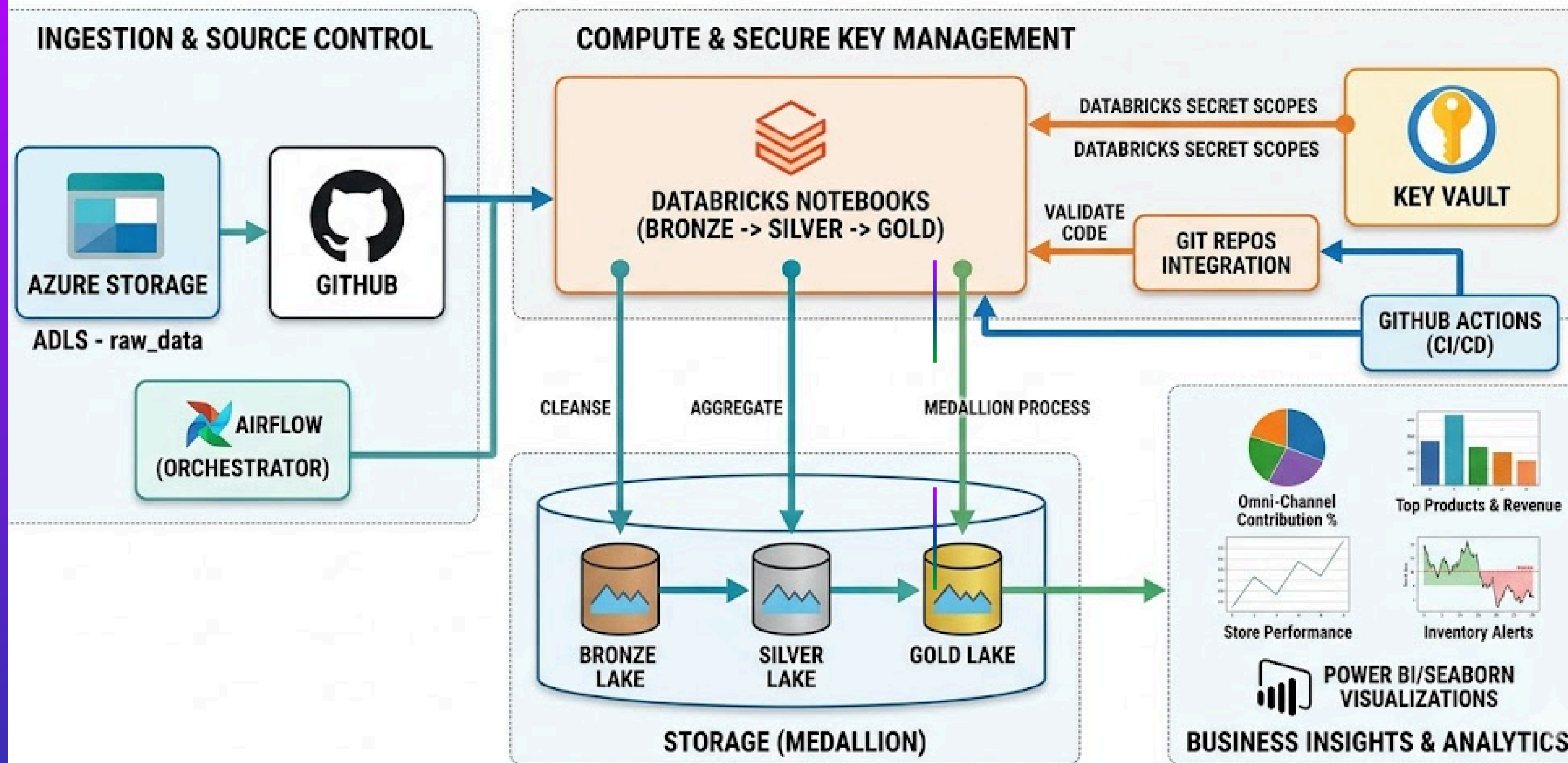
SILVER LAYER: DATA CLEANISING AND SCHEMA VALIDATION



GOLD LAYER: MASTER JOIN & SINGLE VERSION OF TRUTH.



HEXAWARE CAPSTONE RETAIL PIPELINE ARCHITECTURE





**THANK
YOU**